RITHVIK GOGINENI

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EDUCATION

Worcester Polytechnic Institute (WPI) — **Worcester, MA**
B.S. in Mechanical Engineering and Robotics Engineering — *Expected May 2030*

Campion College — **Kingston, Jamaica**
CAPE Sixth Form: Pure Mathematics, Physics, Economics · Second Honours Award (2023–24, 2025–26) — *2019–2026*

ROBOTICS & ENGINEERING

Humanoid Robot | Independent Build — **2025**
Raspberry Pi · ROS · CAD · 3D Printing

- Designed, CAD-modeled, 3D-printed, and assembled a humanoid robot from scratch, integrating structure, electronics, and compute.
- Developed Raspberry Pi/ROS control software and an onboard LLM system that interprets and responds to conversational commands.

FIRST Tech Challenge, Team 16075 | Captain (2025–26), Vice-Captain (2024–25) — **2024–2026**

- Led mechanical integration and iterative development of competition robots (intake, transfer, launching, autonomous), plus competition debugging and engineering documentation.
- Helped earn the **Jamaica Inspire Award** and **Judges' Choice (Ochoa) Award** at the 2025 FIRST Championship; captained the team through the 2025–26 season.

FIRST Global Challenge, Team Jamaica | Lead Engineer — **2025**
Eco-Equilibrium season — *Panama*

- Led robot design, mechanical integration, testing, and competition iteration for Jamaica's national team.
- Finished on the **3rd-place finalist alliance** and placed **3rd for the Katherine Johnson Award** for engineering documentation.

SELECTED PROJECTS

RehabOS (formerly PhysioOS) | Founder & Lead Engineer — **2025–Present**

- Built and deployed an AI-native clinic operating system at Med Rehab Professionals in Kingston, supporting **8+ staff, 100+ patients, and 200+ sessions**.
- Engineered AI-assisted intake with clinical-history conflict detection, room and device tracking, exercise verification, and automated WhatsApp/SMS follow-ups.
- Developed a 3-stage speech pipeline for Jamaican English and Patois; ran Playwright QA across **67 platform pages**.

Roamly | Founder & Developer — **2025–Present**
React Native · TypeScript · Supabase · Mapbox

- Built a navigation app that generates drive, walk, run, and bike routes from mood and time budget, with turn-by-turn voice guidance.
- Migrated routing to Deno/Supabase Edge Functions to isolate API keys and enable cross-user caching.
- Shipped crowdsourced road-safety ratings and an Anthropic-powered multi-day trip planner.

NexusOS (Nexus Inventory) | Founder & Product Architect — **2025–2026**
Next.js · TypeScript · AI

- Designed an AI-native inventory platform for e-commerce SMBs using specialist agents for restocking, forecasting, and suppliers.
- Architected supervisor/critic agent oversight with confidence scoring and human-approval guardrails.

EXPERIENCE & LEADERSHIP

Vaz Preparatory School | Robotics Teacher — **2025–2026**

- Designed and taught hands-on robotics lessons for **30+ students** in Grades 1–6, introducing engineering, programming, and iterative problem-solving.

Halls of Learning / Supreme Ventures Foundation | Robotics Instructor — **2022–2025**

- Advanced from teaching assistant to instructor, teaching **80+ students**, including underserved youth and children in care; mentored teams for national robotics competitions.

Freelance Web Developer & Designer — **2023–Present**

- Built websites and digital branding for Med Rehab Professionals, Proper Constructions Ltd., Gorilla Gym, and Club1962.

Leadership: President, Robotics Club (2025–26; VP 2024–25), leading technical workshops for 30 members · President, Aeronautics Club (2023–26; Secretary 2022–23) · Vice-President, Motorsport Club (2025–26)

TECHNICAL SKILLS

Programming: Python, TypeScript/JavaScript, Java, SQL, C#, Dart
Robotics, Hardware & CAD: ROS, Raspberry Pi, Arduino, REV Robotics, 3D Printing, Fusion 360, Onshape
Software & Infrastructure: React Native, Expo, Flutter, Supabase, PostgreSQL, Edge Functions, Vercel, Mapbox, Playwright, Git/GitHub
AI/ML: OpenAI and Anthropic APIs, LLM agents, speech-to-text pipelines, AI evaluation, multi-armed bandits